ASSINGMENT 9 – Documentation Generation - Automatic documentation and code comments

Task Description :1 – Docstring , Inline comments , Google-style documentation

Task: Compare the three approaches. Discuss their advantages and disadvantages. Explain suitable use cases. Recommend the best documentation style for a mathematical utilities library

Prompt: Explain what the function does, describe the input parameter, and describe the return value. Keep the documentation simple and suitable for a beginner.

Input:

```
def find_max_docstring(numbers):
    """Return the maximum number from a list of numbers."""
    return max(numbers)


def find_max_inline(numbers):
    # Check and return the maximum value in the input list
    return max(numbers)  # Built-in max function handles the comparison


def find_max_google(numbers):
    """Finds the maximum value in a list of numbers.

    Args:
```

```
    numbers (list of int/float): A list of numerical values to search.

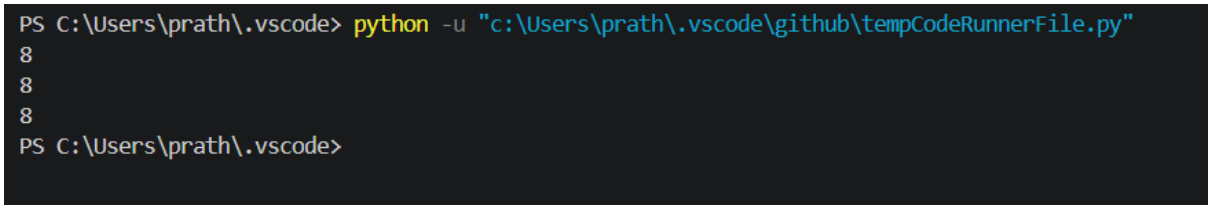
  Returns:
    int/float: The largest number in the list.

  Raises:
    ValueError: If the input list is empty.
  """
  if not numbers:
    raise ValueError("The list of numbers cannot be empty.")
  return max(numbers)
#example usage
numbers = [3, 5, 1, 8, 2]
print(find_max_docstring(numbers))  # Output: 8
print(find_max_inline(numbers))   # Output: 8
print(find_max_google(numbers))   # Output: 8
```

Explanation:

The purpose of this problem is to understand different ways of documenting Python code.

Output:



Task Description :2 – Documentation for login()

Task: Write documentation using all three formats. Critically compare the approaches. Recommend which documentation style would be most useful for new developers joining a project.

Prompt : Generate dd a clear inline comment to the following Python function. The comment should explain what the function is doing without unnecessarily explaining obvious Python syntax.

```
def find_max(numbers):
return max(numbers)
```

Input:

```
def login_docstring(user, password, credentials):
    """Verify if the provided password matches the stored password for the given user in the credentials registry."""
    return credentials.get(user) == password


def login_inline(user, password, credentials):
    # Retrieve password for user and evaluate match
    # Using .get() prevents KeyError if user is missing, returning None which safely compares False
    return credentials.get(user) == password


def login_google(user, password, credentials):
```

```
    """Authenticates a user against a provided credentials store.

    Args:
        user (str): The unique identifier/username of the user trying to log in.
        password (str): The plain-text security password provided by the client.
        credentials (dict): A dictionary mapping usernames (str) to their corresponding correct passwords (str).

    Returns:
        bool: True if the user exists and the password matches; False otherwise.
    """
    return credentials.get(user) == password
#example usage
credentials = {'alice': 'password123', 'bob': 'securepass'}
print(login_docstring('alice', 'password123', credentials))  # Output: True
print(login_inline('alice', 'password123', credentials))   # Output: True
print(login_google('alice', 'password123', credentials))   # Output: True
```

Explanation:

This problem is similar to Problem 1, but the function represents a login/authentication operation.

Output:

```
PS C:\Users\prath\.vscode> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
True
True
True
PS C:\Users\prath\.vscode>
```

Task Description :3 – Calculator Module

Task: Display the module documentation in the terminal. Generate the documentation as an HTML file using Python's pydoc. Open the generated HTML file in a browser and verify it

methods.

Prompt: Generate Convert the following Python function into a properly documented Google-style function. Include a description, Args section, and Returns section.

def find_max(numbers):

return max(numbers)

Input:

"""A basic calculator module providing arithmetic operations with structured documentation.

"""

def add(a, b):

"""Returns the sum of two numbers.

Args:

a (int/float): The first number.

b (int/float): The second number.

Returns:

int/float: The sum of a and b.

"""

return a + b

```
def subtract(a, b):
    """Returns the difference of two numbers.

    Args:
        a (int/float): The first number (minuend).
        b (int/float): The second number (subtrahend).

    Returns:
        int/float: The difference of a and b.
    """
    return a - b


def multiply(a, b):
    """Returns the product of two numbers.

    Args:
        a (int/float): The first factor.
        b (int/float): The second factor.

    Returns:
        int/float: The product of a and b.
    """
    return a * b


def divide(a, b):
```

```
    """Returns the quotient of two numbers.

    Args:
        a (int/float): The dividend.
        b (int/float): The divisor.

    Returns:
        float: The quotient of a divided by b.

    Raises:
        ZeroDivisionError: If the divisor b is zero.
    """
    if b == 0:
        raise ZeroDivisionError("Division by zero is not allowed.")
    return a / b
#example usage
print(divide(10, 2))  # Output: 5.0
print(multiply(3, 4))  # Output: 12
print(subtract(10, 5))  # Output: 5
print(add(7, 3))  # Output: 10
```

Explanation:

This problem introduces automatic documentation generation.

Instead of manually creating a documentation website,Python's pydoc can read the docstrings from your program and generate documentation.

Output:

```
PS C:\Users\prath\.vscode> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
5.0
12
5
10
PS C:\Users\prath\.vscode>
```

Task Description :4 – Conversion Utilities Module

Task: Use Copilot to automatically generate docstrings. Generate the documentation in the terminal. Export the documentation to HTML. Open the HTML documentation in a browser.

Prompt: Generate Compare docstrings, inline comments, and Google-style documentation for the following Python function:

def find_max(numbers):

return max(numbers)

For each approach, explain:

1. Advantages
2. Disadvantages
3. Suitable use cases

Finally, recommend the most effective documentation style for a mathematical utilities library and justify your answer.

Input:

"""A conversion utility module providing functions to convert numbers between decimal, binary, and hexadecimal bases.

"""

def decimal_to_binary(n):

"""Converts a decimal integer to its binary string representation.

```
    Args:
        n (int): The decimal integer to convert.

    Returns:
        str: The binary representation of the integer (prefixed with '0b').

    Raises:
        TypeError: If the input is not an integer.
    """
    if not isinstance(n, int):
        raise TypeError("Input must be an integer.")
    return bin(n)


def binary_to_decimal(b):
    """Converts a binary string to its decimal integer representation.

    Args:
        b (str): The binary string to convert (can optionally start with '0b').

    Returns:
        int: The decimal integer value of the binary string.

    Raises:
        ValueError: If the input string is not a valid binary number.
        TypeError: If the input is not a string.
    """
```

```
    if not isinstance(b, str):
        raise TypeError("Input must be a string.")
    return int(b, 2)


def decimal_to_hexadecimal(n):
    """Converts a decimal integer to its hexadecimal string representation.

    Args:
        n (int): The decimal integer to convert.

    Returns:
        str: The lowercase hexadecimal representation of the integer (prefixed with '0x').

    Raises:
        TypeError: If the input is not an integer.
    """
    if not isinstance(n, int):
        raise TypeError("Input must be an integer.")
    return hex(n)
#example usage
print(decimal_to_binary(10))  # Output: '0b1010'
print(binary_to_decimal('0b1010'))  # Output: 10
print(decimal_to_hexadecimal(255))  # Output: '0xff'
```

Explanation:

This problem combines programming + AI-assisted documentation + automatic documentation generation.

Output:



Task-5: Course Management Module

Prompt: Generate a Python module named course.py for a simple course management system.

Input:

"""A course management module providing utilities to add, remove, and retrieve course details.

"""

In-memory database to store course information

_course_db = {}

def add_course(course_id, name, credits):

  """Adds a new course to the registry.

  Args:

    course_id (str): The unique identifier for the course (e.g., 'CS101').

    name (str): The name of the course.

    credits (int): The number of academic credits for the course.

```
    Raises:
        ValueError: If the course_id already exists in the registry.
        TypeError: If credits is not an integer or if course_id/name is not a string.
    """
    if not isinstance(course_id, str) or not isinstance(name, str):
        raise TypeError("course_id and name must be strings.")
    if not isinstance(credits, int):
        raise TypeError("credits must be an integer.")
    if course_id in _course_db:
        raise ValueError(f"Course with ID {course_id} already exists.")

    _course_db[course_id] = {
        "name": name,
        "credits": credits
    }


def remove_course(course_id):
    """Removes an existing course from the registry.

    Args:
        course_id (str): The unique identifier of the course to be removed.

    Raises:
        KeyError: If the course_id is not found in the registry.
    """
    if course_id not in _course_db:
```

```
        raise KeyError(f"Course with ID {course_id} does not exist.")
    del _course_db[course_id]



def get_course(course_id):
    """Retrieves the details of a specific course.

    Args:
        course_id (str): The unique identifier of the course to retrieve.

    Returns:
        dict: A dictionary containing the course 'name' and 'credits'.

    Raises:
        KeyError: If the course_id is not found in the registry.
    """
    if course_id not in _course_db:
        raise KeyError(f"Course with ID {course_id} does not exist.")
    return _course_db[course_id]
#example usage
add_course('CS101', 'Introduction to Computer Science', 4)
print(get_course('CS101'))  # Output: {'name': 'Introduction to Computer Science', 'credits': 4}
```

Explanation:

Review the course.py module and check whether the documentation for add_course(), remove_course(), and get_course() is complete and consistent.

Output:

```
PS C:\Users\prath\.vscode> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
{'name': 'Introduction to Computer Science', 'credits': 4}
PS C:\Users\prath\.vscode>
```